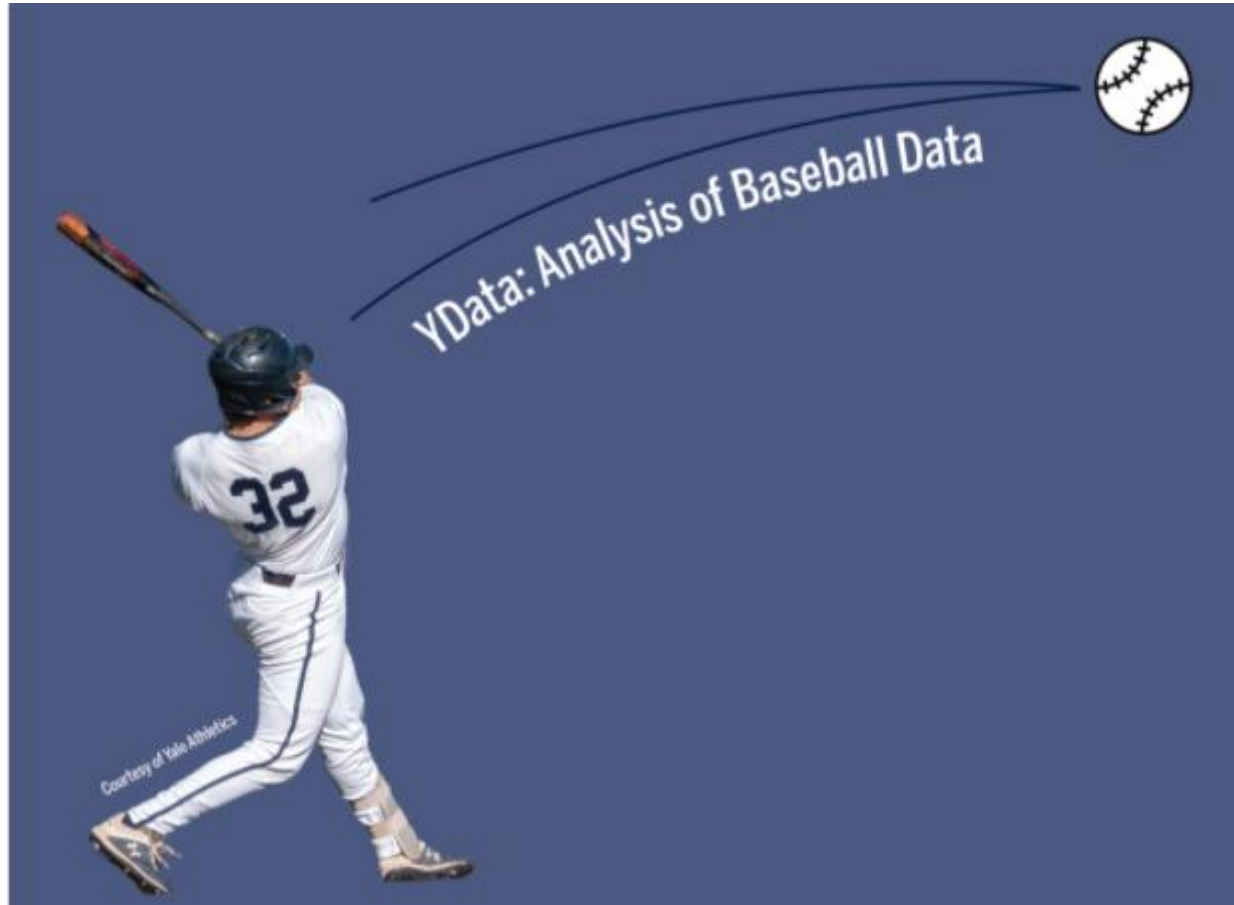


S&DS 1730



Class 5: Additional pandas operations and interactive graphics

Overview

Brief discussion of Astrobball

Discussing when does the home/visiting team win a baseball

A few additional pandas operations and data visualizations

- Pivot tables
- Heatmaps
- Converting between long and wide data formats

If there is time: Interactive graphics

- Jupyter notebook widgets
- Plotly

A conversation with Sean Ahmed, Principal Analyst for the Detroit Tigers

Announcement: Lab 4

Lab 4 has been released. It is due on
Sunday February 15th

How is did lab 3 go?



Lab 4: Estimate the probability of winning a game

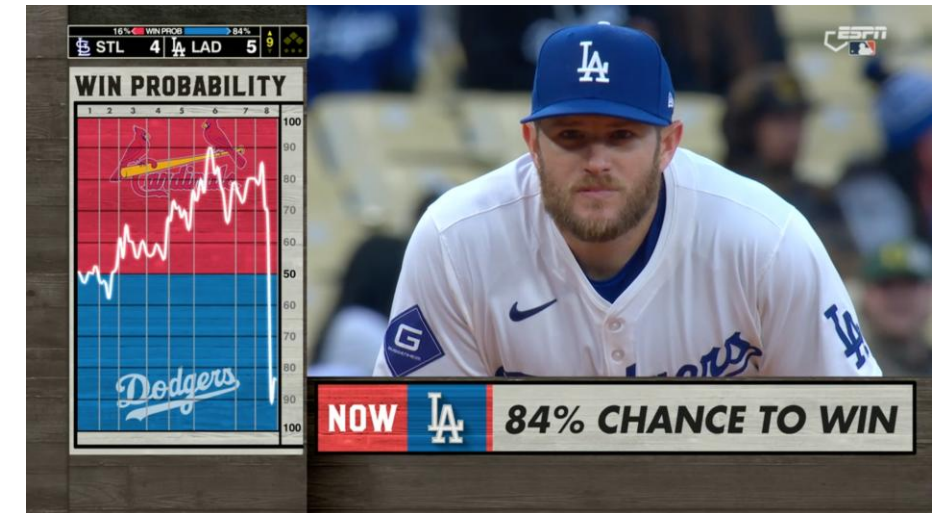
In lab 4 you will calculate the probability of winning a game given the score difference and the inning/number of outs

- Also apply these probability to specific games to see how the probability changes over a game

Note: probability is a number between 0 and 1

- Probability of 0 means an event will not happen
- Probability of 1 means an event will definitely happen
- Probability between 0 and 1 is the likelihood that an outcome will happen given what is known
 - $P(X = x \mid \text{context})$
 - $P(\text{Home team wins} \mid \text{score, inning, outs})$

We will discuss probability more in next two weeks

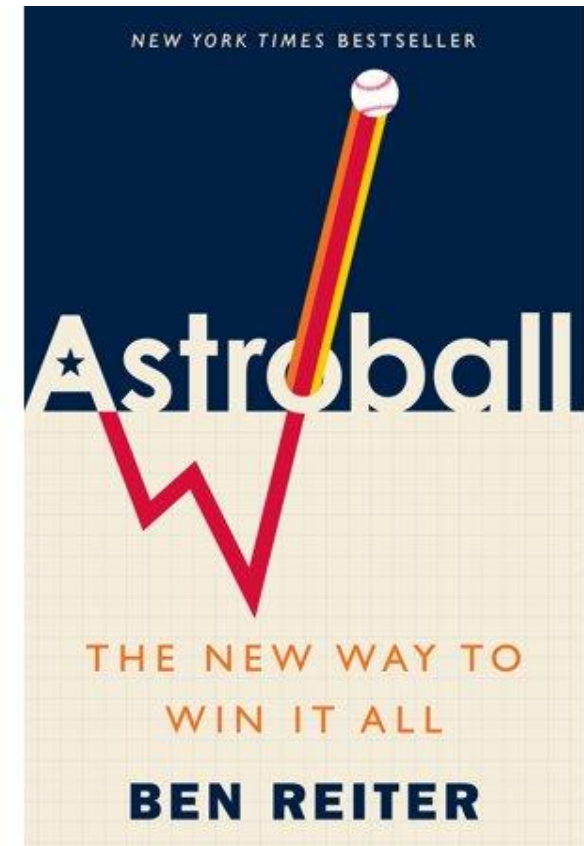


Recommendation: get started early!

Discussion on chapter 3 of Astrobball

Let's discuss the chapter 3 for 3 minutes in small groups of 3-4 people and then have a larger conversation as a group

- E.g., Discuss your quote and reaction to chapter 3

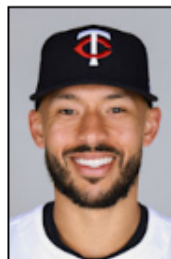


Thoughts on the chapter 3 of Astrobball?

Drafting Correa 2012

Correa projected statistics, pg 70

- BA: .272-.287
- HR: 35-40 HR



Carlos Correa

Positions: Shortstop and Third Baseman

Bats: Right • Throws: Right

6-3, 220lb (190cm, 99kg)

Team: [Houston Astros](#) (majors)

[More bio, uniform, draft, salary info ▼](#)

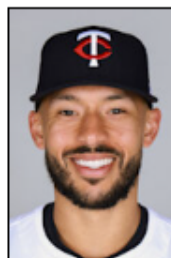
SUMMARY	WAR	AB	H	HR	BA	R	RBI	SB	OBP	SLG	OPS	OPS+
2025	1.5	537	148	13	.276	63	52	0	.332	.402	.734	103
Career	45.7	4716	1298	200	.275	686	724	33	.352	.464	.816	124

Season	Age	Team	Lg	WAR	G	PA	AB	R	H	2B	3B	HR	RBI	SB	CS	BB	SO	BA	OBP	SLG	OPS	OPS+	rOBA	Rbat+	TB	GIDP	HBP	SH	SF	IBB	Pos	Awards
2015	20	HOU	AL	4.7	99	432	387	52	108	22	1	22	68	14	4	40	78	.279	.345	.512	.857	135	.371	136	198	10	1	0	4	2	6	MVP-24,ROY-1
2016	21	HOU	AL	7.0	153	660	577	76	158	36	3	20	96	13	3	75	139	.274	.361	.451	.811	124	.372	138	260	12	5	0	3	5	*6/H	
2017 ★	22	HOU	AL	6.7	109	481	422	82	133	25	1	24	84	2	1	53	92	.315	.391	.550	.941	155	.409	164	232	12	2	0	4	5	6/D	AS,MVP-17
2018	23	HOU	AL	2.9	110	468	402	60	96	20	1	15	65	3	0	53	111	.239	.323	.405	.728	99	.316	94	163	17	2	0	11	3	*6/D	
2019	24	HOU	AL	3.8	75	321	280	42	78	16	1	21	59	1	0	35	75	.279	.358	.568	.926	137	.385	136	159	8	2	0	4	0	6	
2020	25	HOU	AL	1.6	58	221	201	22	53	9	0	5	25	0	0	16	49	.264	.326	.383	.709	93	.324	98	77	4	3	0	1	2	*6/H	
2021 ★	26	HOU	AL	7.3	148	640	555	104	155	34	1	26	92	0	0	75	116	.279	.366	.485	.850	131	.369	133	269	16	4	0	6	2	*6	AS,MVP-5,GG
2022	27	MIN	AL	5.3	136	590	522	70	152	24	1	22	64	0	1	61	121	.291	.366	.467	.834	138	.367	140	244	18	3	0	4	2	*6/DH	
2023	28	MIN	AL	1.3	135	580	514	60	118	29	2	18	65	0	0	59	131	.230	.312	.399	.711	94	.312	93	205	30	4	0	3	3	*6	
2024 ★	29	MIN	AL	3.7	86	367	319	55	99	20	2	14	54	0	0	40	61	.310	.388	.517	.905	152	.393	155	165	5	3	0	4	4	6/DH	AS
2025	30	2TM	AL	1.5	144	584	537	63	148	29	0	13	52	0	0	45	113	.276	.332	.402	.734	103	.323	103	216	19	1	0	1	0	65/H	
2025	30	MIN	AL	0.1	93	364	337	40	90	19	0	7	31	0	0	26	68	.267	.319	.386	.704	94	.308	91	130	12	0	0	1	0	6/H	
2025	30	HOU	AL	1.3	51	220	200	23	58	10	0	6	21	0	0	19	45	.290	.355	.430	.785	117	.349	123	86	7	1	0	0	0	5/6	
11 Yrs				45.7	1253	5344	4716	686	1298	264	13	200	724	33	9	552	1086	.275	.352	.464	.816	124	.358	126	2188	151	30	0	45	28	*65/HD	
162 Game Avg				5.9	162	691	610	89	168	34	2	26	94	4	1	71	140	.275	.352	.464	.816	124	.358	126	283	20	4	0	6	4		

Drafting Correa 2012

Correa projected statistics, pg 70

- BA: .272-.287
- HR: 35-40 HR



Carlos Correa

Positions: Shortstop and Third Baseman

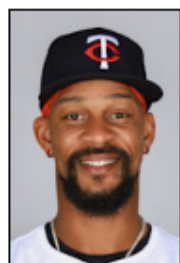
Bats: Right • Throws: Right

6-3, 220lb (190cm, 99kg)

Team: [Houston Astros](#) (majors)

[More bio, uniform, draft, salary info ▼](#)

SUMMARY	WAR	AB	H	HR	BA	R	RBI	SB	OBP	SLG	OPS	OPS+
2025	1.5	537	148	13	.276	63	52	0	.332	.402	.734	103
Career	45.7	4716	1298	200	.275	686	724	33	.352	.464	.816	124



Byron Buxton

Position: Centerfielder

Bats: Right • Throws: Right

6-2, 190lb (188cm, 86kg)

Team: [Minnesota Twins](#) (majors)

[More bio, uniform, draft, salary info ▼](#)

SUMMARY	WAR	AB	H	HR	BA	R	RBI	SB	OBP	SLG	OPS	OPS+
2025	4.9	488	129	35	.264	97	83	24	.327	.551	.878	136
Career	29.8	3102	768	168	.248	523	436	117	.308	.487	.795	114



Kevin Gausman

Position: Pitcher

Bats: Left • Throws: Right

6-2, 205lb (188cm, 92kg)

Team: [Toronto Blue Jays](#) (majors)

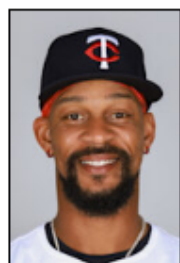
[More bio, uniform, draft, salary info ▼](#)

SUMMARY	WAR	W	L	ERA	G	GS	SV	IP	SO	WHIP
2025	3.8	10	11	3.59	32	32	0	193.0	189	1.062
Career	27.8	112	113	3.81	361	322	0	1911.0	1954	1.240

Drafting Correa 2012

More info on the 2012 draft

- [Wikipedia](#)
- [Baseball reference](#)



Byron Buxton

Position: Centerfielder
Bats: Right • Throws: Right
6-2, 190lb (188cm, 86kg)
Team: [Minnesota Twins](#) (majors)

[More bio, uniform, draft, salary info ▼](#)

SUMMARY	WAR	AB	H	HR	BA	R	RBI	SB	OBP	SLG	OPS	OPS+
2025	4.9	488	129	35	.264	97	83	24	.327	.551	.878	136
Career	29.8	3102	768	168	.248	523	436	117	.308	.487	.795	114



Enos Cabell

Positions: Third Baseman, First Baseman and Outfielder
Bats: Right • Throws: Right
6-4, 170lb (193cm, 77kg)
Born: [October 8, 1949](#) (Age: 71-142d) in Fort Riley, [KS](#) 

[More bio, uniform, draft, salary info ▼](#)

SUMMARY	WAR	AB	H	HR	BA	R	RBI	SB	OBP	SLG	OPS	OPS+
Career	11.3	5952	1647	60	.277	753	596	238	.308	.370	.678	93



Kevin Gausman

Position: Pitcher
Bats: Left • Throws: Right
6-2, 205lb (188cm, 92kg)
Team: [Toronto Blue Jays](#) (majors)

[More bio, uniform, draft, salary info ▼](#)

SUMMARY	WAR	W	L	ERA	G	GS	SV	IP	SO	WHIP
2025	3.8	10	11	3.59	32	32	0	193.0	189	1.062
Career	27.8	112	113	3.81	361	322	0	1911.0	1954	1.240

Trusting a model...

"It's one thing to get somebody with college-level statistics to create a model" Sig said.

"It's another for the decision-makers to really use it.

I don't mean use it as a tiebreaker, or to throw the analyst a bone late in the draft.

I mean really *use* it, from the first pick.

That's what Jeff did."

Brief note: When does a team win a game?

Brief note: When does a team win a game?

1. The **visiting team wins** if they are winning 9th inning (or later inning) and they get the home team to have 3 outs in the bottom of the 9th (or later) inning.
2. The **home team wins** if they are winning in the top of the 9th inning and they get the visiting team to have 3 outs.
 - In this case the home team does not bat in the bottom of the 9th
3. The **home team** has a "walk-off win" if they are tied or losing in the bottom of the 9th (or later inning) and they score more runs than the visiting team
 - In this case the game ends before the home team has 3 outs.

	1	2	3	4	5	6	7	8	9	10		R	H	E
VISITOR	0	0	0	0	0	0	0	2	0			2	0	0
HOME	0	0	0	0	0	0	0	0	3			3	0	0

If there is rain, the game can be stopped if 5 innings have been played

Pivot tables and heat maps

Pivot Tables and heatmaps

Pivot tables aggregate values based on two grouping variables, and create a table where:

- The rows are the levels of one cat. variable
- The columns are the levels of the second cat. variable
- The values are aggregated over a third quant. variable

```
df2 = df.pivot_table(index = "col1", columns = "col2",  
                      values = "col3", aggfunc = "max")
```

```
nba2 = nba.pivot_table(index = "TEAM", columns = "POSITION",  
                       values = "SALARY", aggfunc = "max")
```

rows columns values

	PLAYER	TEAM	POSITION	SALARY
0	De'Andre Hunter	Atlanta Hawks	SF	9.835881
1	Jalen Johnson	Atlanta Hawks	SF	2.792640
2	AJ Griffin	Atlanta Hawks	SF	3.536160
3	Trent Forrest	Atlanta Hawks	SG	0.508891
4	John Collins	Atlanta Hawks	PF	23.500000

	POSITION	C	PF	PG	SF	SG
TEAM						
Atlanta Hawks		18.206896	23.500000	37.096500	9.835881	17.071120
Boston Celtics		26.500000	4.306281	22.900000	30.351780	17.142857
Brooklyn Nets		9.391069	44.119845	38.917057	20.100000	19.500000
Charlotte Bobcats		3.722040	1.563518	8.623920	30.075000	21.486316
Chicago Bulls		3.200000	7.775400	9.030000	27.300000	37.096500

Pivot Tables and heatmaps

One can then visualize the data as a heatmap using seaborn

```
sns.heatmap(nba2) # seaborn
```

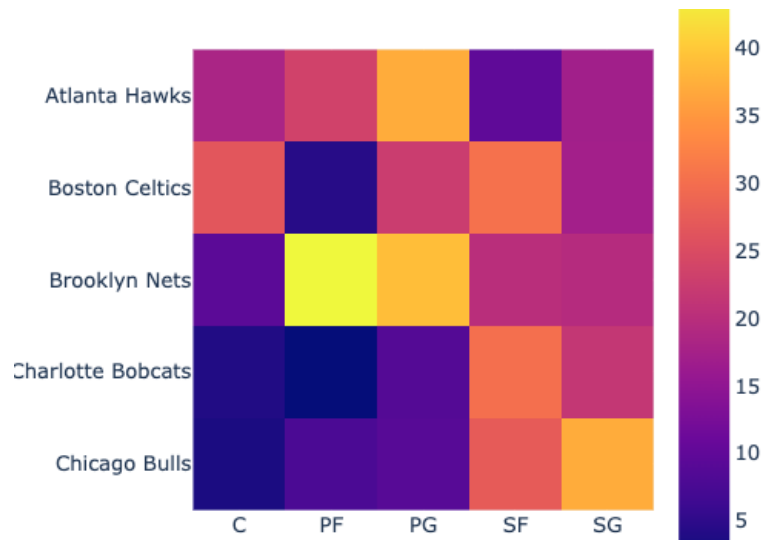


Diagram illustrating the relationship between a data table and its pivot table:

The top table (original data) has columns: **PLAYER**, **TEAM**, **POSITION**, and **SALARY**. Rows are labeled 0 to 4. Red boxes highlight the **TEAM**, **POSITION**, and **SALARY** columns. Arrows labeled "rows", "columns", and "values" point to these respective components.

The bottom table (pivot table) has columns: **POSITION**, **C**, **PF**, **PG**, **SF**, and **SG**. The rows are labeled by **TEAM**. Red arrows connect the pivot table columns to the corresponding columns in the original data table.

	PLAYER	TEAM	POSITION	SALARY
0	De'Andre Hunter	Atlanta Hawks	SF	9.835881
1	Jalen Johnson	Atlanta Hawks	SF	2.792640
2	AJ Griffin	Atlanta Hawks	SF	3.536160
3	Trent Forrest	Atlanta Hawks	SG	0.508891
4	John Collins	Atlanta Hawks	PF	23.500000

	POSITION	C	PF	PG	SF	SG
	TEAM					
	Atlanta Hawks	18.206896	23.500000	37.096500	9.835881	17.071120
	Boston Celtics	26.500000	4.306281	22.900000	30.351780	17.142857
	Brooklyn Nets	9.391069	44.119845	38.917057	20.100000	19.500000
	Charlotte Bobcats	3.722040	1.563518	8.623920	30.075000	21.486316
	Chicago Bulls	3.200000	7.775400	9.030000	27.300000	37.096500

Let's explore this in Jupyter!

Long and wide data formats

Wide vs. Long data

Data can be held in a DataFrame in different formats:

- **Wide format:** Many columns and fewer rows
- **Long format:** Many rows and fewer columns

Wide data

Person	Age	Height
Bob	32	72
Alice	24	65
Steve	64	70

Long data

Person	name	value
Bob	Age	32
Bob	Height	72
Alice	Age	24
Alice	Height	65
Steve	Age	64
Steve	Height	70

Converting from wide to long format

To convert a DataFrame from **wide** to **long** we can use `my_df.melt()`

- **id_vars**: Columns that should stay intact (and be duplicated)
- **var_name**: Name of the new column that will store the *former column names*
- **value_name**: Name of the new column that will store the *values* from the wide columns

`my_df.melt(`

`id_vars = "Person",`

`var_name = "name",`

`value_name = "value")`

Wide data

Person	Age	Height
Bob	32	72
Alice	24	65
Steve	64	70



Long data

Person	name	value
Bob	Age	32
Bob	Height	72
Alice	Age	24
Alice	Height	65
Steve	Age	64
Steve	Height	70

Converting from long to wide format

To convert a DataFrame from **long** to **wide** we can use `my_df.pivot()`

- **index:** Columns that should be the new Index column
- **columns:** A column of strings that will become the *new column names*
- **values:** The column whose values that will be filled in

Long data

Person	name	value
Bob	Age	32
Bob	Height	72
Alice	Age	24
Alice	Height	65
Steve	Age	64
Steve	Height	70



Wide data

Person	Age	Height
Bob	32	72
Alice	24	65
Steve	64	70

```
my_df.pivot(  
    index = "Person",  
    columns = "name",  
    values = "value")
```

Let's try it in Jupyter!

Jupyter widgets

Jupyter widgets

Jupyter widgets allow you to add buttons, sliders, etc. to a Jupyter notebook so that you can create interactive visualizations

We can add widgets by importing:

```
import ipywidgets as widgets
```

We can then create a widget using:

```
slider = widgets.IntSlider(value=10)
```

```
# Create a slider  
slider = widgets.IntSlider(value=10)  
slider
```



```
slider.value
```

15

Jupyter widgets

There are several ways to connect widgets to figures

- One way is through `widgets.interact()`

Boolean argument: checkbox

String argument: textbox

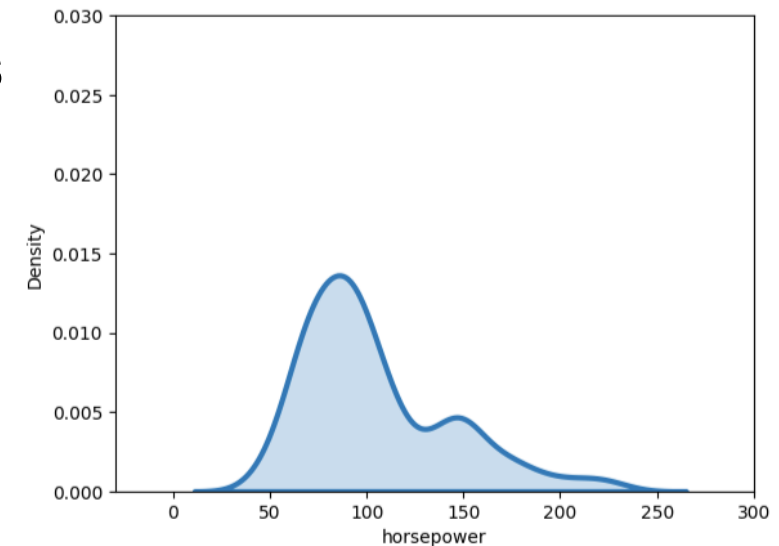
List argument: dropdown menu

Numeric arguments: slider

```
def bandwidth_widget(bw = 1):  
    sns.kdeplot(cars.horsepower,  
                bw_adjust=bw)
```

```
widgets.interact(bandwidth_widget, bw = (.1, 3));
```

This slider updates the figure



Overrides default arguments



Let's try it in Jupyter!

Interactive graphics with Plotly

Interactive visualizations for data exploration

Interactive visualizations are useful for exploring data to find trends

- They can be shared on the internet
- They can't be put in static pdfs

We will use plotly to create interactive graphics

```
import plotly.express as px
```



Plotly interactive plots

Line plots

```
fig = px.line(data_frame = , x = , y = , color = , hover_name = , line_shape = )
```

Scatter plots

```
fig = px.scatter(data_frame = , x = , y = , size = , color = , hover_name = )
```

Add axis labels

```
fig.update_layout(xaxis_title="X", yaxis_title="Y")
```

Let's explore this in Jupyter!